

STAGE 1

Leng et al. 2021
AD snRNA-seq
(entorhinal cortex, n=42)

AD-vulnerable
RORB+ excitatory neurons
(Braak stage $\geq$ IV)

486 DEGs
(Wilcoxon $FDR < 0.05$)

STAGE 2

Neftel et al. 2019
GBM snRNA-seq
(adult tumors, n=28)

Neural-mimicry cells
NPC-like + OPC-like
(malignant filter applied)

1,576 DEGs
(Wilcoxon $FDR < 0.05$)

STAGE 3

Cross-disease convergence testing

3 independent tests:
• RRHO2 (rank-rank hypergeometric overlap)
• Hypergeometric enrichment • Permutation (10^4)

H2 REJECTED
(1/3 tests passed
pre-specified criteria)

Post-hoc discovery:
DU quadrant (18 genes)
DD quadrant (20 genes)

STAGE 4

Independent cohort validation (DD signature)

AD cohort:
GSE125583 (n=289)
fusiform gyrus
Braak-stratified

Dose-response
Wilcoxon $p = 1.3 \times 10^{-5}$

GBM cohort:
TCGA-GBM (n=157) vs
GTEx cortex (n=510)
recount3 pipeline

Cross-disease convergence
Spearman $\rho = 0.713$
($p = 4.2 \times 10^{-4}$)